

Kimberly Wang

San Jose, CA / Troy, NY |  | [linkedin.com/in/wangkimb/](https://www.linkedin.com/in/wangkimb/) | kimbwang.github.io

EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. Electrical Engineering, GPA: 3.71

Aug 2024 – Expected Dec 2027

Relevant Coursework: Electric Circuits, Computer Components and Operations, Embedded Control, Intro to Logic Based AI, Intro to Electronics (Spring 2026), Intro to Engineering Design (Spring 2026), Signals and Systems (Spring 2026)

TECHNICAL SKILLS

Hardware: Arduino, MSP432, Basys 3, Soldering, Oscilloscope, Waveform Generator, Spectrum Analyzer, Multimeter

Software: LTspice, Cadence, Altium Designer, Vivado, SOLIDWORKS, LabVIEW, MATLAB, Simulink, Office Suite

Languages: Python, C, VHDL, MATLAB

EXTRACURRICULARS

Rensselaer Motorsport Member (FSAE)

Dec 2025 – Present

Rensselaer Polytechnic Institute

Troy, NY

- Team member of the team responsible for low voltage components of the EV
- Contribute to circuit and PCB design using SOLIDWORKS for different applications
- Responsible for soldering, assembling, and testing electronics to be integrated into the EV

Undergraduate Student Assistant for "Intro to ECSE"

Aug 2025 – Dec 2025

Rensselaer Polytechnic Institute

Troy, NY

- Help students in debugging and implementing C programming on Arduino and MATLAB
- Advise on analog and digital circuit building, testing, and measurement
- Clarify questions about theoretical concepts within contexts of labs

Undergraduate Researcher @ RPI Mercer XLab

Jan 2025 – Present

Rensselaer Polytechnic Institute

Troy, NY

Software Configured Breadboard (SCB) Supervised by ADI Fellow Douglas Mercer

- Used Python and ALICE software to computerize the wiring of breadboards to reduce debugging and wiring issues
- Executed board-level validation and root-cause analysis utilizing automated Python scripts and a multimeter
- Authored technical documentation and user guides to streamline the onboarding process for new users
- Collaborating with RPI's AI club to utilize Cursor AI to automate circuit design, building, and testing

PROJECTS

CO Sensor System | LTspice, Oscilloscope, Waveform Generator, Spectrum Analyzer

Sep 2025 – Present

- Created a multi-stage analog system for maintaining safety from CO in buildings
- Designed and built the initial indication stage using an RC filter and comparator with an LED output
- Configured autonomous fan controls to trigger active ventilation controls upon the indication signal
- Utilized first and second order filters and 555 Timers to achieve desired signal inputs
- Prototype final stage of an audible alerting sub-system for added safety

LED Sequencer | Altium Designer

Nov 2025

- Designed a custom PCB in Altium Designer following manufacturing constraints on trace widths and clearance
- Generated manufacturing deliverables (Gerbers, BOM) and selected components based on vendor specifications

Compass Controlled TI RSLK Car | MSP432, C, I2C

Apr 2025

- Designed a closed-loop navigation system using a magnetometer via I2C protocols for real time directions
- Implemented PID controls in C to minimize errors in motor movements
- Configured GPIO interrupts to handle data acquisition and maintain stable motor control

PI Controller Design | LTspice, Oscilloscope, Spectrum Analyzer

Nov 2024

- Designed and simulated an analog PI controller in LTspice to achieve a steady state voltage in 4s
- Verified the design by building and testing the circuit to get results with < 5% error from simulation